

Reference Link:

<https://community.sap.com/t5/technology-blog-posts-by-members/a-complete-tutorial-on-odata-based-adobe-forms-for-sap-purchase-orders/ba-p/13754410>

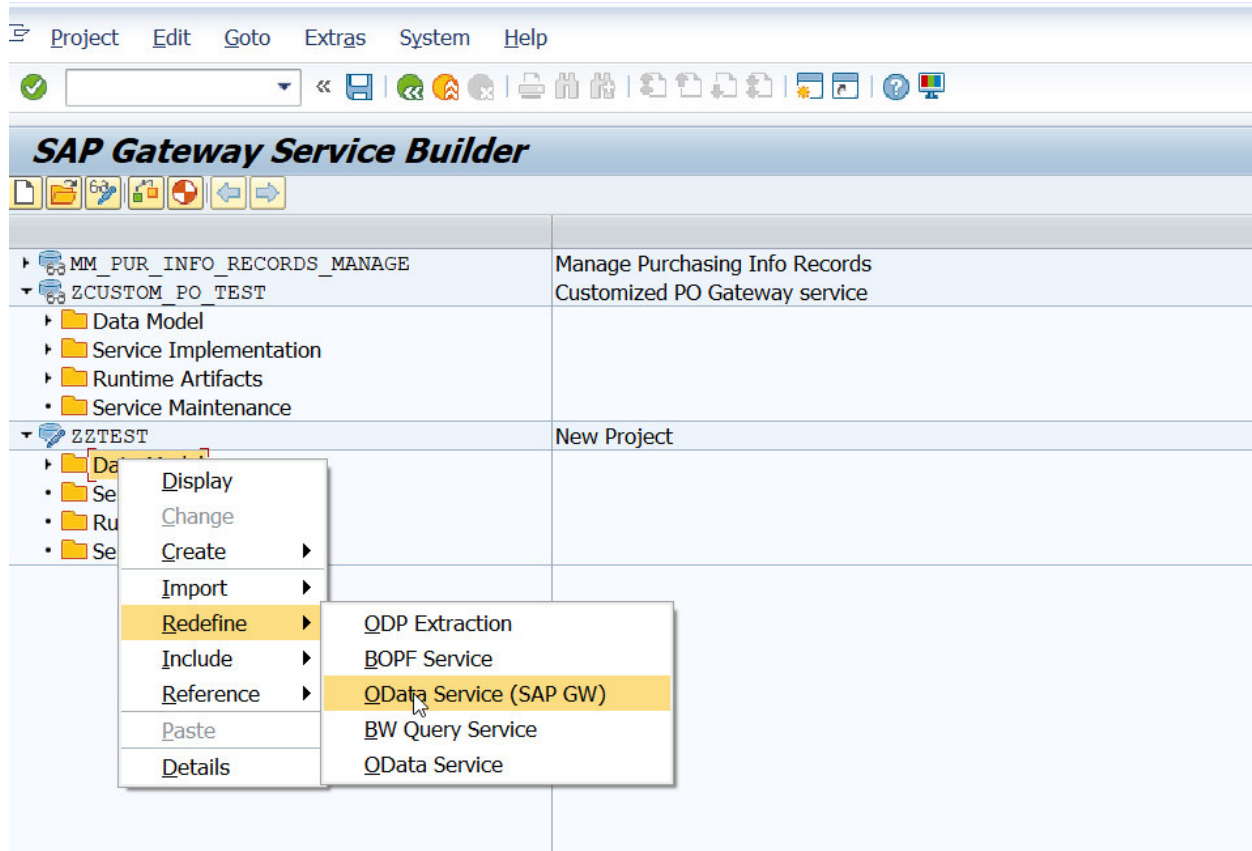
Started off creating a new Adobe form by copying the ZTEST_PUR_PURCHASE_ORDER

The screenshot displays the Adobe Form Builder application window. The title bar reads "Form Builder: Change Form ZTEST_PUR_PURCHASE_ORDER". The interface includes a menu bar (Form, Edit, Layout, Utilities, Environment, System, Help) and a toolbar. The "Properties" tab is active, showing the following details:

- Description:** Form Template Purchase Order. This form uses an XML schema-based interface.
- Interface:** SOMU_DEFAULT_INTERFACE
- Gateway Service:** ZCUSTOM_PO_TEST_SRV
- Fragment Usage:** B Form with fragment
- Country/Region List:** (Empty field)
- Fonts in PDF Output:** Do not embed fonts
- Mirror Layout:** No mirroring in RTL languages
- Optimize Data:** Not optimized
- Created By:** US5092
- Changed By:** US5092
- Date:** 08/05/2026
- Time:** 16:00:16
- Package:** Z001
- Original Language:** EN

Next I created the Gateway service.

Open SEGW> Create New project>



Adobe Form Development Process for SAP Systems.

The Adobe form development process in SAP typically involves three main steps:

1. Creating the Interface
2. Designing the Form
3. Data Binding

While the overall process remains consistent, the interface creation can vary depending on the data retrieval method. The three primary approaches are:

1. DDIC (Data Dictionary) based forms
2. XML Schema based forms
3. Gateway based forms

Let's focus on enhancing a gateway-based form using a Purchase Order example:

Approach:

Enhancing a Gateway-Based Purchase Order Form.

In this scenario, we'll add two additional fields to the standard form:

1. Bill to Address
2. Amount in Words

which is discussed in detail in the later steps below:

A. Search for Form in Maintain Form Template App:

Step 1: Locate the standard Purchase Order form.

- Access Fiori App "Maintain Form Template"
- Search for "MM_PUR_PURCHASE_ORDER" under "Form Template Name".
- Find the template under "Predelivered Template" category.

Standard * ▾ Hide Filter Bar Filters (1)

Form Template Name: =MM_PUR... x Description: Data Source: Available Languages: Available Countries / Regions:

Form Type: ▾

Custom Templates **Predelivered Templates**

Predelivered Templates (1) Copy [F] ⚙

Form Template Name	Description	Available Languages	Available Countries / Regions
MM_PUR_PURCHASE_ORDER	Form Template Purchase Order	DE, EN	>

Step 2: Create a copy of the standard form.

- Select "MM_PUR_PURCHASE_ORDER" under "Predelivered Templates"
- Click on the copy option
- Enter a new name for the custom form template
- Click OK to create the copy

Output Management Form Template

Standard * ▾

Form Template Name: Description:

Form Type:

Available Countries / Regions:

Custom Templates **Predelivered Templates**

Predelivered Templates (1)

Form Template Name	Description	Available Languages	Available Countries / Regions
MM_PUR_PURCHASE_ORDER	Form Template Purchase Order	DE, EN	

Copy Form Template

Source Template: MM_PUR_PURCHASE_ORDER

Prefix: ZZ1_

Form Template Name: *

OK Cancel

Copy [≡] [⚙]

Step 3: Verify the new custom form

- Insert the new form name "ZZ1_HM_TEST" in the "Form Template Name" field
- Go to the "Custom Templates" tab
- Review the details of the newly created custom form

Output Management Form Template

Standard * ▾

Form Template Name: Description: Data Source: Available Languages: Available Countries / Regions:

Form Type:

Custom Templates **Predelivered Templates**

Custom Templates (1)

Form Template Name	Description	Available Languages	Available Countries / Regions
ZZ1_HM_TEST	Form Template Purchase Order	DE, EN	

New Delete Copy [≡] [⚙]

Step 4: Examine form template details

- Click on the form template to open it.
- Review the detailed information of the custom form.

< **SAP** Output Management Form Template ▾

Search ⓘ Bell User

Output Management Form Template

☆ Save as Tile

Standard ▾ Hide Filter Bar Filters (1)

Form Template Name: Description: Data Source: Available Languages: Available Countries / Regions:

Form Type:

Custom Templates Predelivered Templates

Custom Templates (1) New Delete Copy [≡] ⚙

Form Template Name	Description	Available Languages	Available Countries / Regions
Content			
☒ ZZ1_HM_TEST	Form Template Purchase Order	DE, EN	>

Step 5: Identify the OData service.

- Look for "Data Source" section in the form details.
- The OData service name "FDP_EF_PURCHASE_ORDER_SRV" will appear for Purchase Order.

< **SAP** Output Management Form Template ▾

Search ⓘ Bell User

ZZ1_HM_TEST

Form Template Purchase Order ☆ Save as Tile

Details

Form Template Name: * ZZ1_HM_TEST

Description: Form Template Purchase Order

Data Source: * FDP_EF_PURCHASE_ORDER_SRV ▾

Form Type: * Content ▾

Embed Fonts in PDF: ☐

Mirror Layout in RTL languages: ☐

Original Language: EN

Original Language:

Available Countries / Regions: ▾

Now, as we have created the custom form template “ZZ1_HM_TEST”, we will go for creation of custom OData service to provide the data source to the adobe form which is explained in the next step.

B. Custom OData Service Creation (SEGW):

Step 1: Create OData Service Project

- Go to transaction code SEGW
- Create a new project named "ZHM_PO_TEST"

SAP Gateway Service Builder

Create Project

Project: ZHM_PO_TEST

Description: Test Purchase Order GW Service

Attributes

Project Type: Service with SAP Annotations

Generation Strategy: Standard

Object Directory Entry

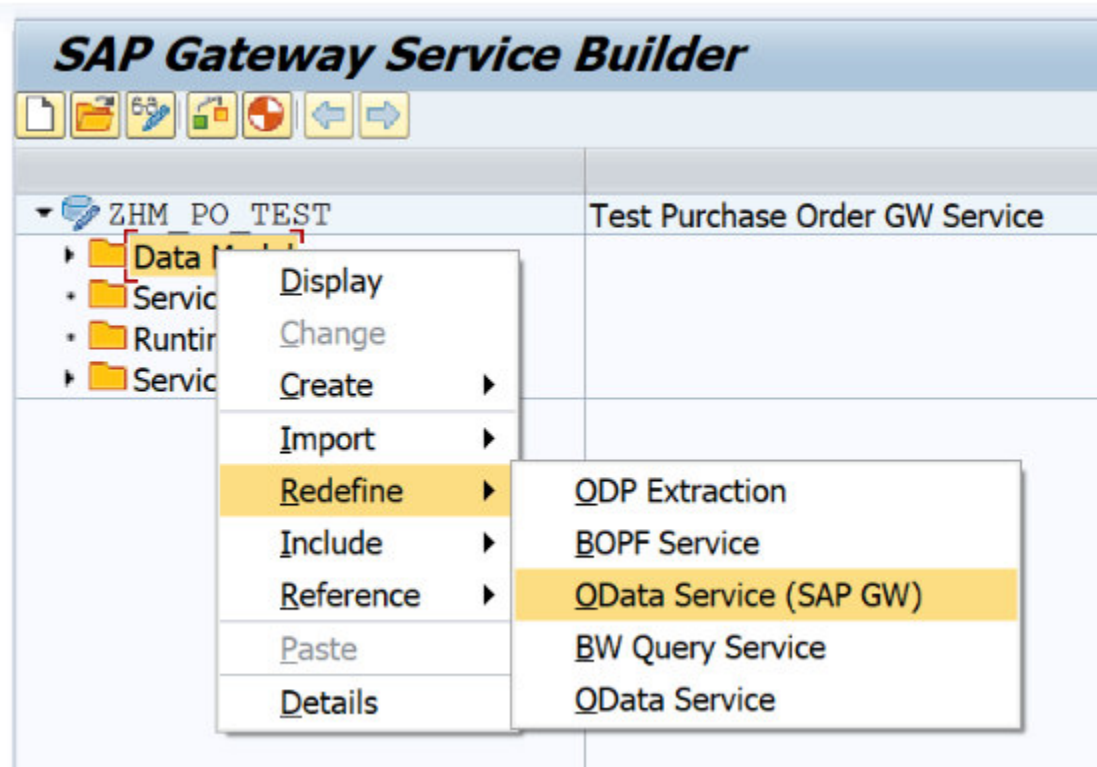
Package: [Redacted]

Person Responsible: [Redacted]

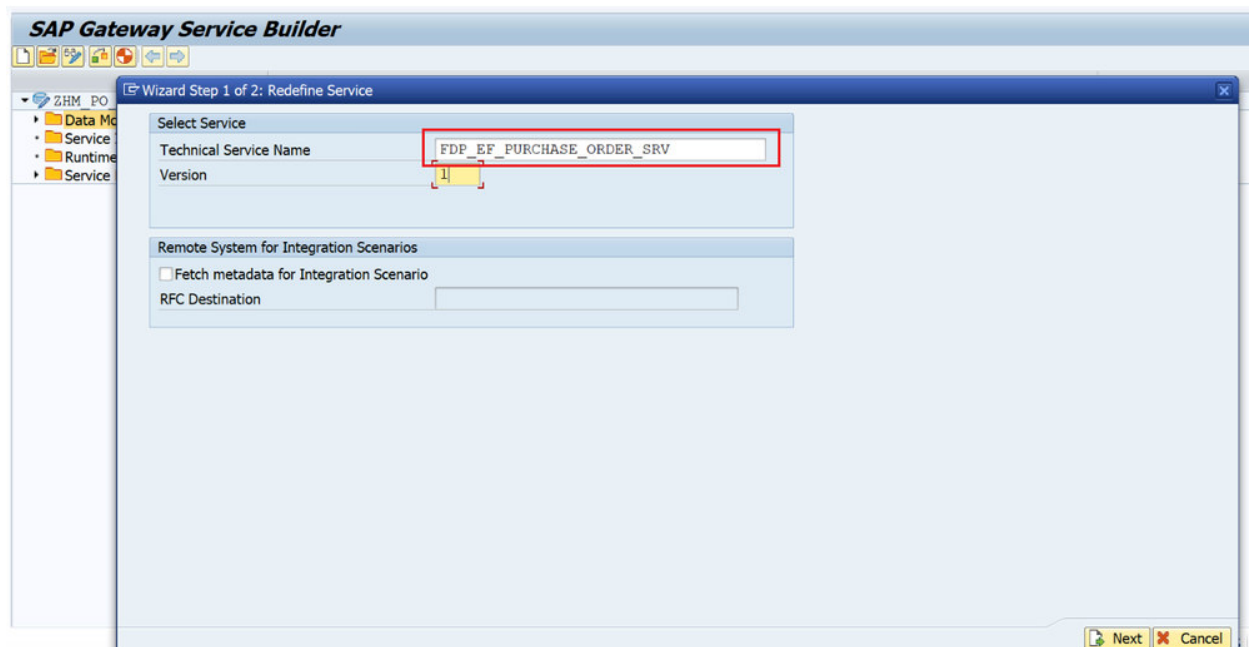
Local Object

Step 2: Redefine Data Model

- In project "ZHM_PO_TEST", Right-click on "Data Model" -> "Redefine" -> "OData Service (SAP GW)".



- Enter technical service name: "FDP_EF_PURCHASE_ORDER_SRV"
- Specify Version: "0001" and Next.
- Select all fields and Finish.



Step 3: Generate Runtime Artifacts

- Select "Generate Runtime Artifacts"
- Check the box for "Overwrite Base/Extended Service" and Press Enter

SAP Gateway Service Builder

Model and Service Definition

Model Provider Class

Class Name: ZCL_ZHM_PO_TEST_MPC_EXT

Base Class Name: ZCL_ZHM_PO_TEST_MPC

Data Provider Class

☒ Generate Classes

Class Name: ZCL_ZHM_PO_TEST_DPC_EXT

Base Class Name: ZCL_ZHM_PO_TEST_DPC

Service Registration

Technical Model Name: ZHM_PO_TEST_MDL

Model Version: 1

Technical Service Name: ZHM_PO_TEST_SRV

Service Version: 1

Service Extension

☒ Overwrite Base/Extended Service

! Base service will be inaccessible!

Technical Service Name: FDP_EF_PURCHASE_ORDER_SRV

Service Version: 1

Step 4 : Identify and Add the Custom Fields.

In this requirement, "Bill To Address" and "Amount In Words" are needed to be add, as addition fields in the form apart from the standard details. So, for that we need to check which details are available in the OData Service.

- Bill To Address: (Not found in Standard OData Service, but it's a custom field)

Since, Bill to Address is not there in any of the entity in OData service, we will add this, in the structure of Purchase Order Node, i.e., TDS_ME_PO_HEADER.

- Amount in Words: (Not found in Standard OData Service, but it's a custom field)

Since, we have the standard field for fetching the Total Amount and now, we just need to convert it in Words. For that we will add the custom field in the structure of Purchase Order Node, i.e., TDS_ME_PO_HEADER.

SAP Gateway Service Builder

Entity Types

Name	ABAP Structure	Base Type	Abstr...	Label	La... S
PurchaseOrderHierItemNode	TDS_ME_PO_ITEM		☐		T
PurchaseOrderHierItemSTTe...			☐		T
PurchaseOrderHierItemTexts			☐		T
PurchaseOrderHierScheduleL...	TDS_ME_PO_SCHEDULELI...		☐		T
PurchaseOrderItemBatchNode	TDS_ME_PO_ITEM_BATCH		☐		T
PurchaseOrderItemChanges...			☐		T
PurchaseOrderItemManufact...	TDS_ME_PO_MPN		☐		T
PurchaseOrderItemNode	TDS_ME_PO_ITEM		☐		T
PurchaseOrderItemSTTexts			☐		T
PurchaseOrderItemTexts			☐		T
PurchaseOrderLimitItemCha...			☐		T
PurchaseOrderLimitItemNode			☐		T
PurchaseOrderLimitItemSTT...			☐		T
PurchaseOrderLimitItemTexts			☐		T
PurchaseOrderNode	TDS_ME_PO_HEADER		☐		T

Messages

- Runtime objects for project 'ZHM_PO_TEST' were generated successfully
 - Model Provider Base Class 'ZCL_ZHM_PO_TEST_MPC' generated successfully
 - Model Provider Extension Class 'ZCL_ZHM_PO_TEST_MPC_EXT' generated successfully
 - Data Provider Base Class ZCL_ZHM_PO_TEST_DPC generated successfully
 - Data Provider Implementation Class ZCL_ZHM_PO_TEST_DPC_EXT generated successfully
 - Data provider base class ZCL_ZHM_PO_TEST_DPC implemented successfully
 - Service ZHM_PO_TEST_SRV was registered successfully

Step 5: Create Append Structure

- Go to transaction SE11
- Create a new Append Structure named "ZPO_BILL_ADDRESS"

Dictionary: Display Structure

Structure: TDS_ME_PO_HEADER Active

Short Description: PO Purchase Order Node Header Structure

Attributes Components Input Help/Check Currency/quantity fields

Appends for TDS_ME_PO_HEADER

Object Name: Status Short text

Create Append Structure for TDS_ME_PO_HEADER

Append Name: ZPO_BILL_ADDRESS

ADDRESS LINE 4 Types CHAR 80 0 0

Step 6: Add and Activate Custom Fields

- Add the required fields and activate the append structure

Dictionary: Display Append Structure

Append Structure: ZPO_BILL_ADDRESS Active

Short Description: Bill Address Details

Attributes Components Input Help/Check Currency/quantity fields

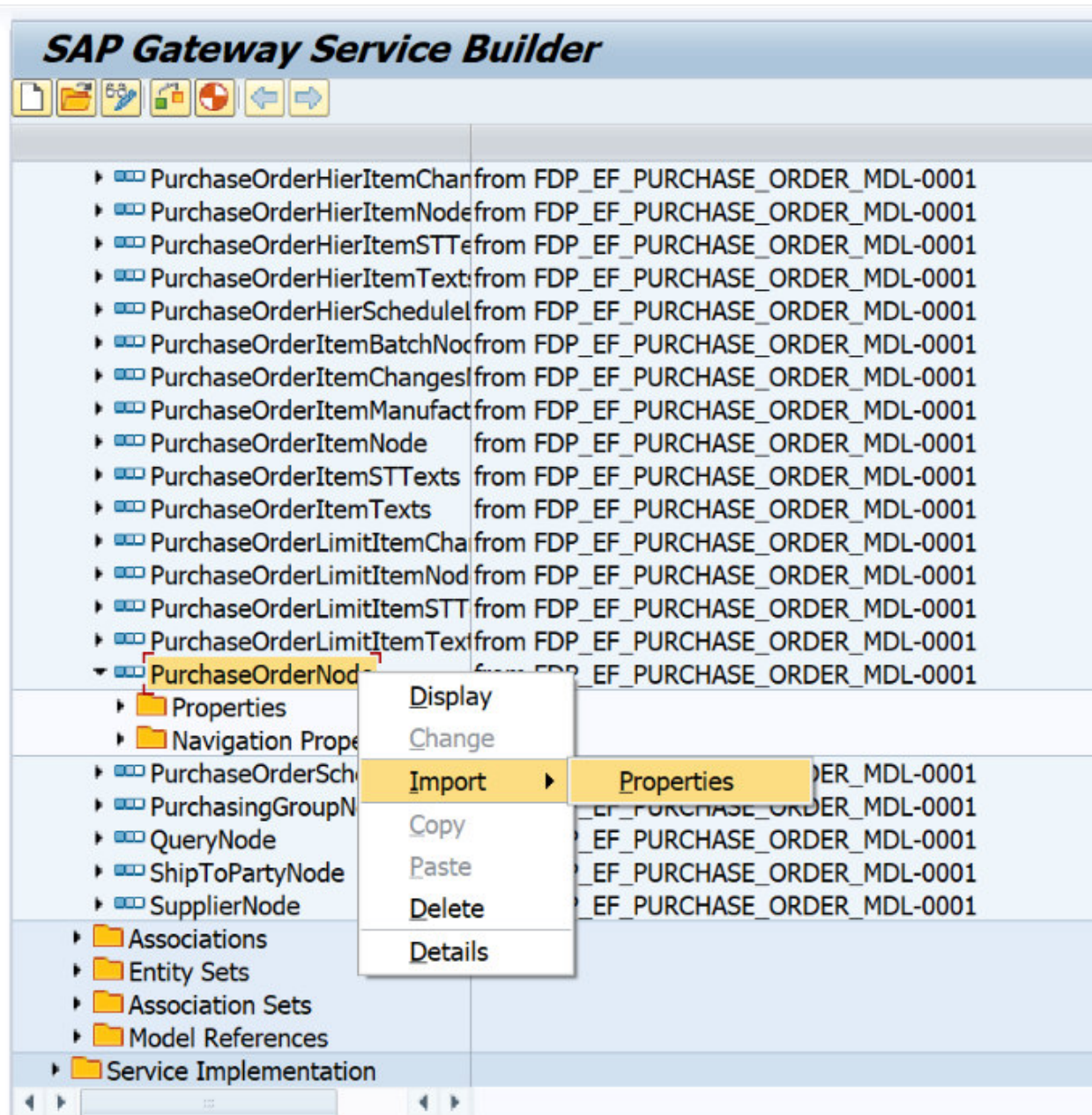
Built-In Type Show Appending Obj 1 / 10

Component	Typing Method	Component Type	Data Type	Length	Decim...	Coordinate	Short Description
BILL ADD LINE 1	Types	CHAR80	CHAR	80	0		0 Char 80
BILL ADD LINE 2	Types	CHAR80	CHAR	80	0		0 Char 80
BILL ADD LINE 3	Types	CHAR80	CHAR	80	0		0 Char 80
BILL ADD LINE 4	Types	CHAR80	CHAR	80	0		0 Char 80
BILL ADD LINE 5	Types	CHAR80	CHAR	80	0		0 Char 80
BILL ADD LINE 6	Types	CHAR80	CHAR	80	0		0 Char 80
BILL ADD LINE 7	Types	CHAR80	CHAR	80	0		0 Char 80
BILL ADD LINE 8	Types	CHAR80	CHAR	80	0		0 Char 80
SALESPERSON	Types	EVERK	CHAR	30	0		0 Responsible Salesperson at Supplier's Office
ZWORD AMOUNT	Types	IN WORDS	CHAR	510	0		0 Amount in words

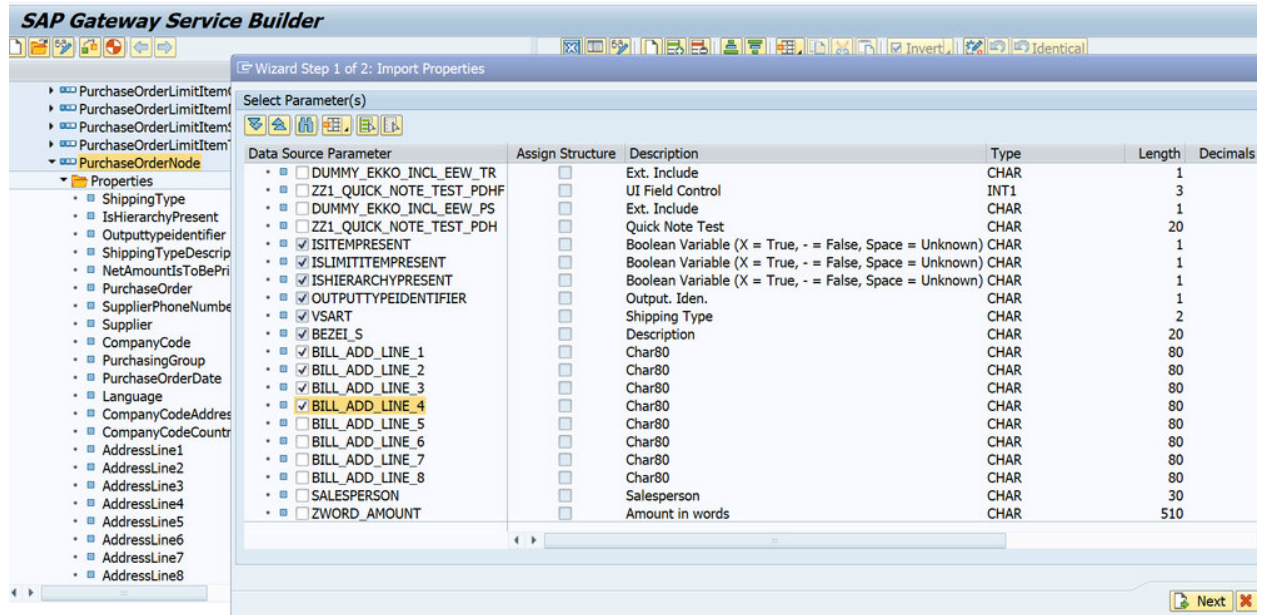
Step 7: Import New Properties to Entity

- Return to SEGW transaction
- Open project "ZHM_TEST"
- Locate Entity "PurchaseOrderNode"

- Right-click on "PurchaseOrderNode" -> Select "Import" -> "Properties"

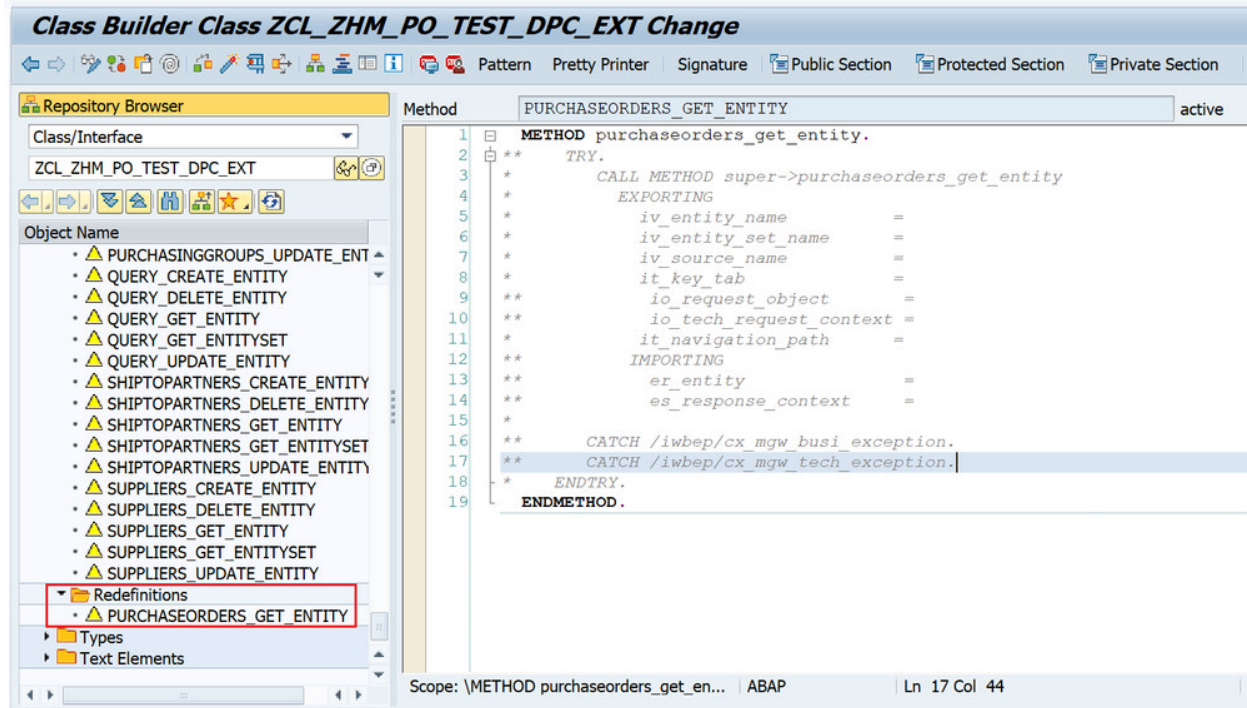


- And add additional field in "PurchaseOrderNode", which were added in the structure "TDS_ME_PO_HEADER" in the prior step.



Step 8: Update Runtime Artifacts

- Expand the "Service Implementation" node.
- Locate and expand the "Runtime Artifacts" subfolder.
- Find the class "ZCL_ZHM_PO_TEST_DPC_EXT" in the list.
- In the class, look for the method "PURCHASEORDER_GET_ENTITY" and redefine it, by right click on method -> Redefine.



Step 9: Code for fetching Bill to Address and Amount in Words.

METHOD purchaseorders_get_entity.

TRY.

CALL METHOD super->purchaseorders_get_entity

EXPORTING

iv_entity_name = iv_entity_name

iv_entity_set_name = iv_entity_set_name

iv_source_name = iv_source_name

it_key_tab = it_key_tab

io_request_object = io_request_object

io_tech_request_context = io_tech_request_context

it_navigation_path = it_navigation_path

IMPORTING

er_entity = er_entity

es_response_context = es_response_context.

CATCH /iwbep/cx_mgw_busi_exception.

CATCH /iwbep/cx_mgw_tech_exception.

ENDTRY.

DATA: lv_spell_words TYPE spell,

lv_amount TYPE p DECIMALS 2,

lv_amt_spell(255) TYPE c.

"Bill To Address

SELECT SINGLE

FROM i_purchaseorder

FIELDS purchaseorder, _purchaseorderitem-plant,

_purchaseorderitem_plant-addressid AS addressid, _Supplier-

Country AS suppliercountry,

_Supplier_SupplierPurchasingOrg-SupplierRespSalesPersonName AS salesperson

WHERE purchaseorder = @er_entity-ebeln

INTO @DATA(ls_data).

IF sy-subrc = 0.

IF ls_data IS NOT INITIAL AND ls_data-suppliercountry IS NOT INITIAL.

cl_fdp_ef_pur_ord_form_utility=>get_address_in_printform(

EXPORTING

iv_language = er_entity-spras

iv_sender_country = ls_data-suppliercountry

iv_adrn = ls_data-addressid

iv_street_has_priority = abap_true

IMPORTING

ev_address_line1 = er_entity-bill_add_line1

ev_address_line2 = er_entity-bill_add_line2

ev_address_line3 = er_entity-bill_add_line3

ev_address_line4 = er_entity-bill_add_line4

).

ENDIF.

"Sales person

er_entity-salesperson = ls_data-salesperson.

"Amount in Words

lv_amount = er_entity-purchaseordernetamount.

CALL FUNCTION 'SPELL_AMOUNT'

EXPORTING

amount = lv_amount

currency = er_entity-waers

filler = ''

language = sy-langu

IMPORTING

```
in_words = lv_spell_words
```

```
EXCEPTIONS
```

```
not_found = 1
```

```
too_large = 2
```

```
OTHERS = 3.
```

```
IF sy-subrc <> 0.
```

```
* Implement suitable error handling here
```

```
ENDIF.
```

```
IF lv_spell_words IS NOT INITIAL.
```

```
er_entity-zword_amount = lv_spell_words-word.
```

```
ENDIF.
```

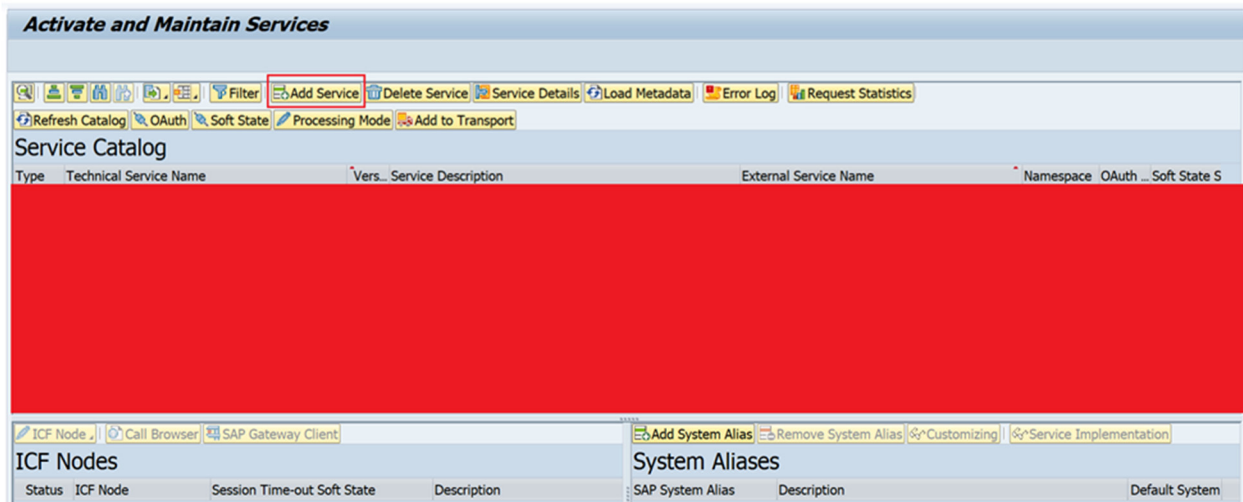
```
ENDIF.
```

C. Register the service in GW (T-code - /IWFND/MAINT_SERVICE)

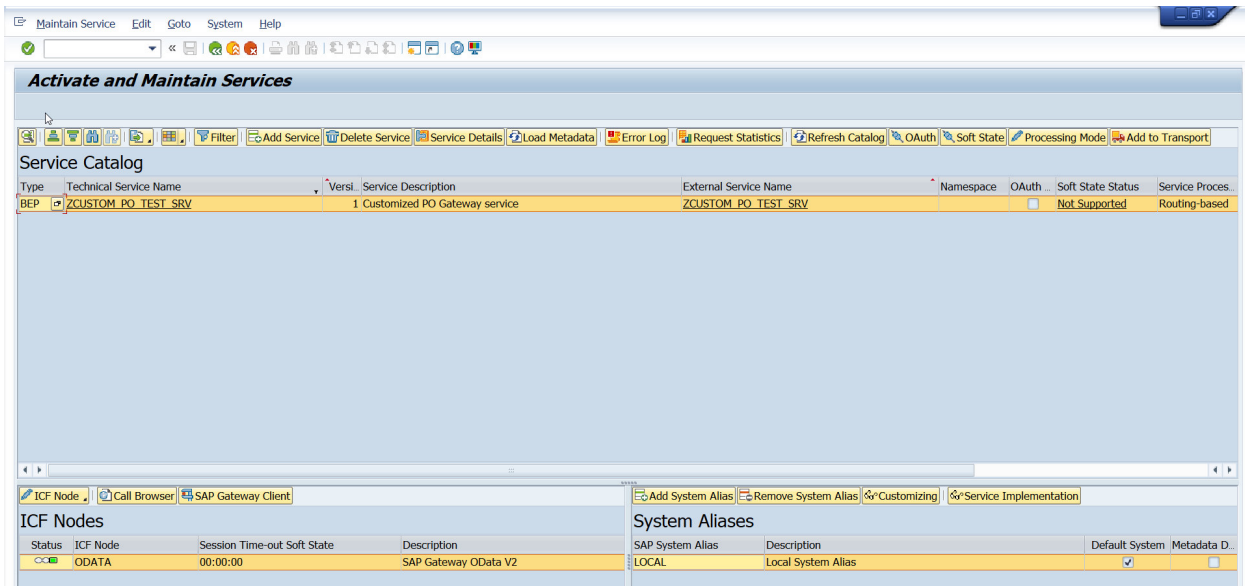
Post creation of OData service, registration of the service will be done by following below steps:

Step 1 : Adding the service in GW.

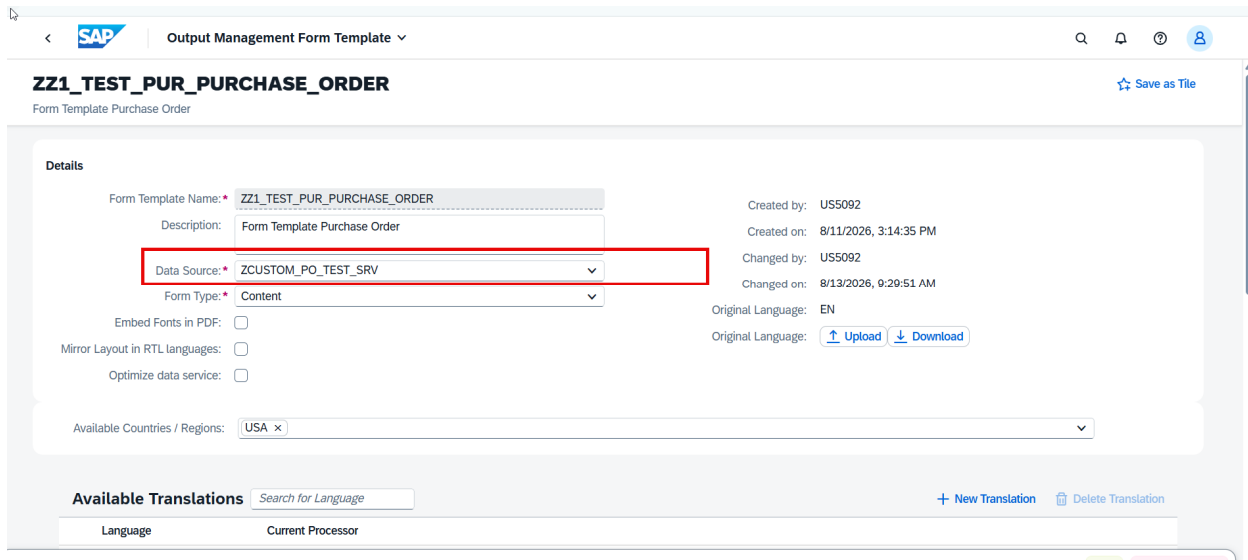
- Click on “Add Service” button



Now your service is successfully registered.



Step 2 : Change the ODATA service using MaintainForm Template app, If the service is unavailable from the Maintain form, Then use SFP and change it. It should appear here. This must be done before downloading the form to make any changes.



E. Download the bindings for the form

There are two ways to download the bindings for the forms.

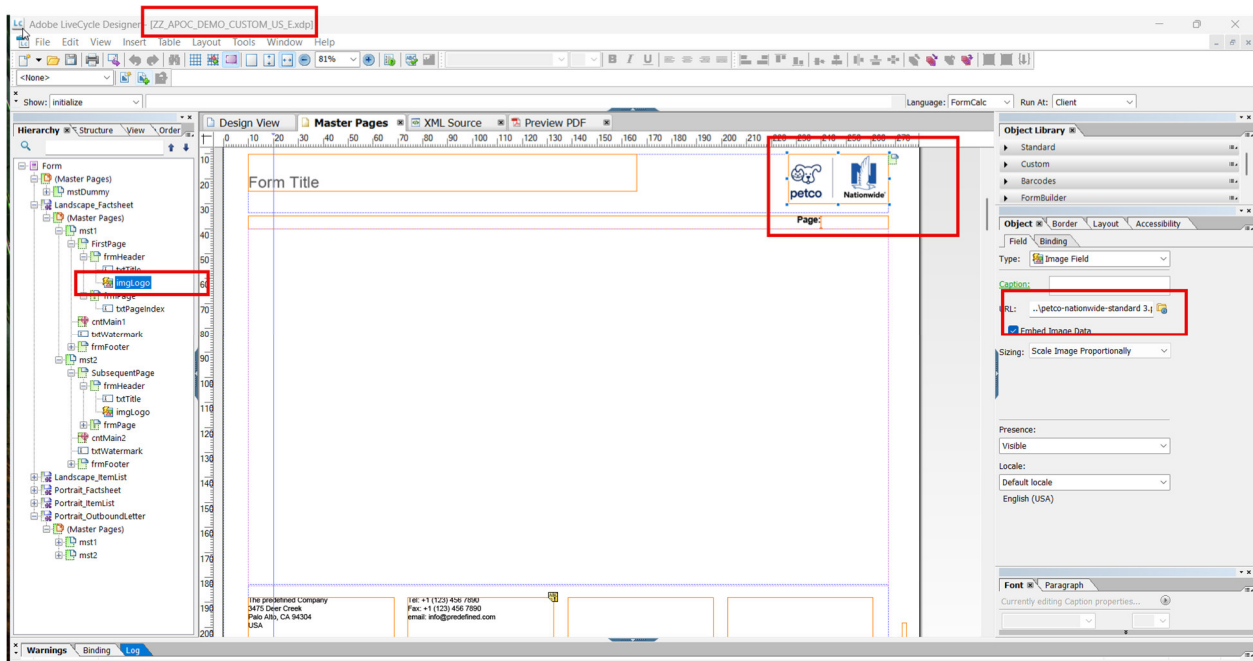
Bindings can be downloaded from:

RESTRICTED

1) First and foremost download copy the Master Template form that you want to use. In this case I made a copy of APOC_DEMO_CUSTOM_US to ZZ_APOC_DEMO_CUSTOM_US

2) Maintain Form Template App -> Custom Form Name -> Download (it will download both XDP and XSD file in the folder). Download the Master Form.

3) Open the Master form in the Adobe Life Cycle Design using Open File. I changed the LOGO to use Petco and Nationwide to various formats of the master form.



Save the file. Go to SFP > Upload Layout and then use the .xsd file from the location to upload the changes. Activate the form ZZ_APOC_DEMO_CUSTOM_US in SFP.

Now Let's work on the Content form to add in the new fields and bind them to the new ODATA service.

4) Maintain Form Template App -> Custom Form Name -> Download (it will download both XDP and XSD file in the folder). Download the Content form and use the Master Form that was created

RESTRICTED

Form Template Name: ZZ1_TEST_PUR_PURCHASE_ORDER
Description: Form Template Purchase Order
Data Source: ZCUSTOM_PO_TEST_SRV
Form Type: Content
Embed Fonts in PDF: ☐
Mirror Layout in RTL languages: ☐
Optimize data service: ☐
Available Countries / Regions: USA x

Download Template ZZ1_TEST_PUR_PURCHASE_ORDER (EN)
Select Form Template Master:
ZZ_APOC_DEMO_CUSTOM_US
☐ Download without Form Template Master
OK Cancel

Open the form in LCD and switch to the right Master Form Fragment to be used.

Form Title

Purchase Order:

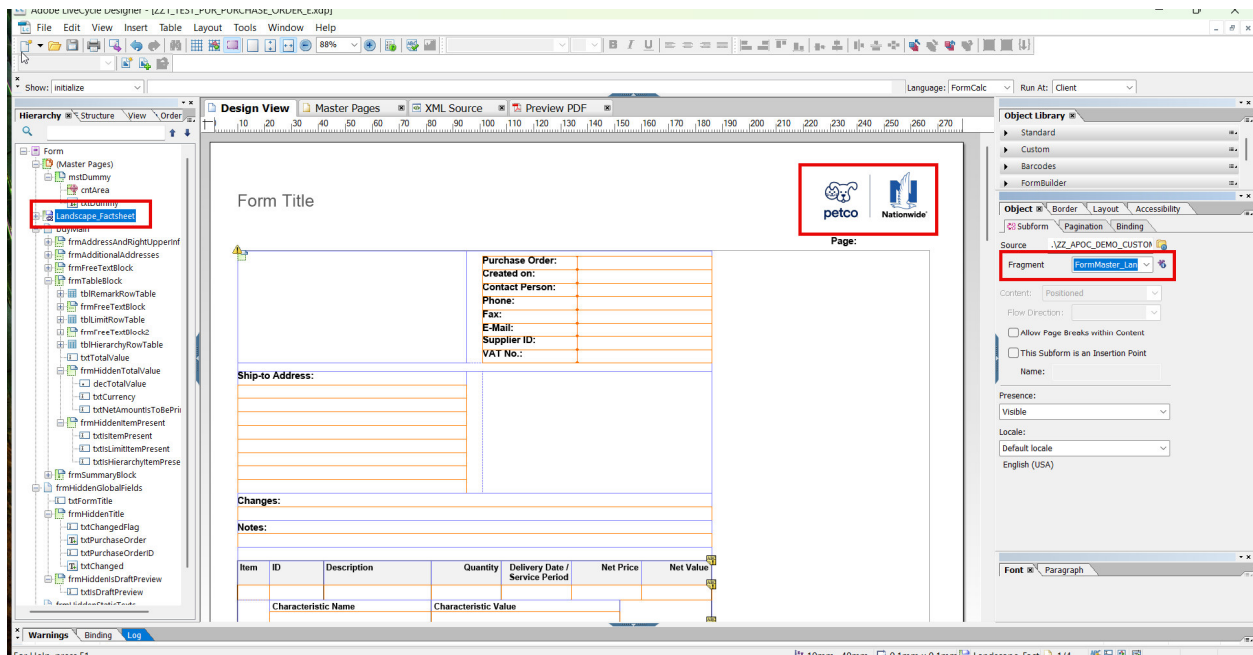
Created on:	
Contact Person:	
Phone:	
Fax:	
E-Mail:	
Supplier ID:	
VAT No.:	

Ship-to Address:

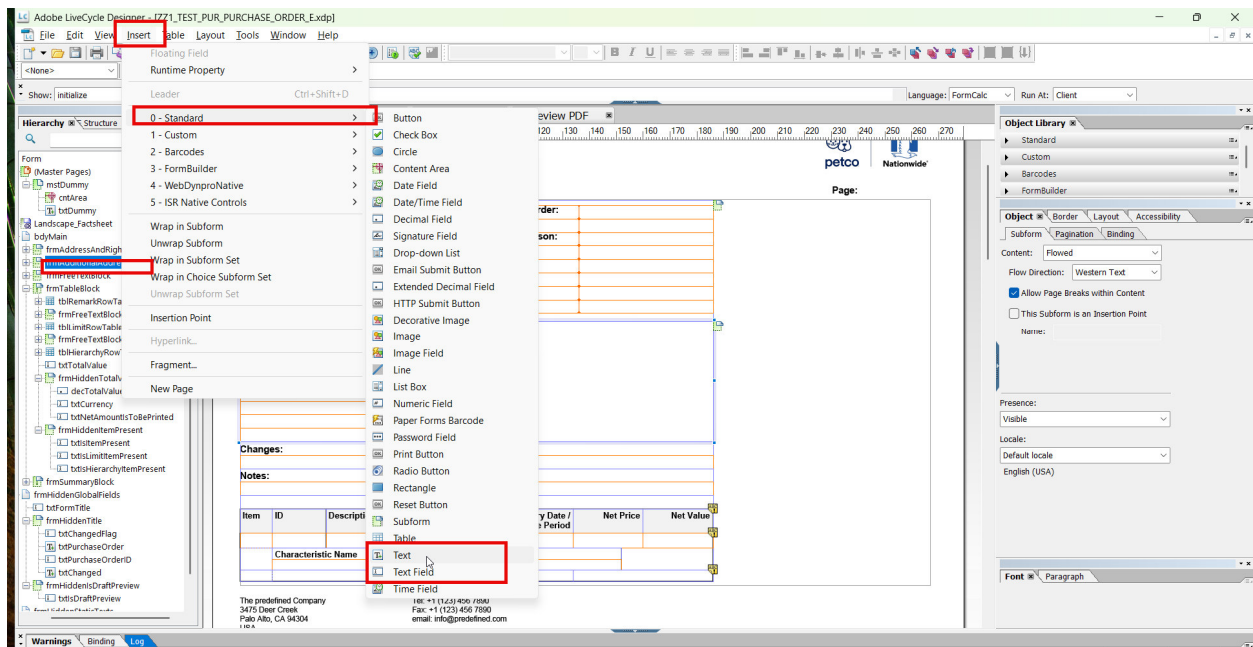
Changes:

Navigate to the landscape fragment and select the right master fragment to use. In this case the one with the new Petco logo. SAVE.

RESTRICTED



Now to add new fields, Locate the form where you need to be and click insert.



New text field was added.

RESTRICTED

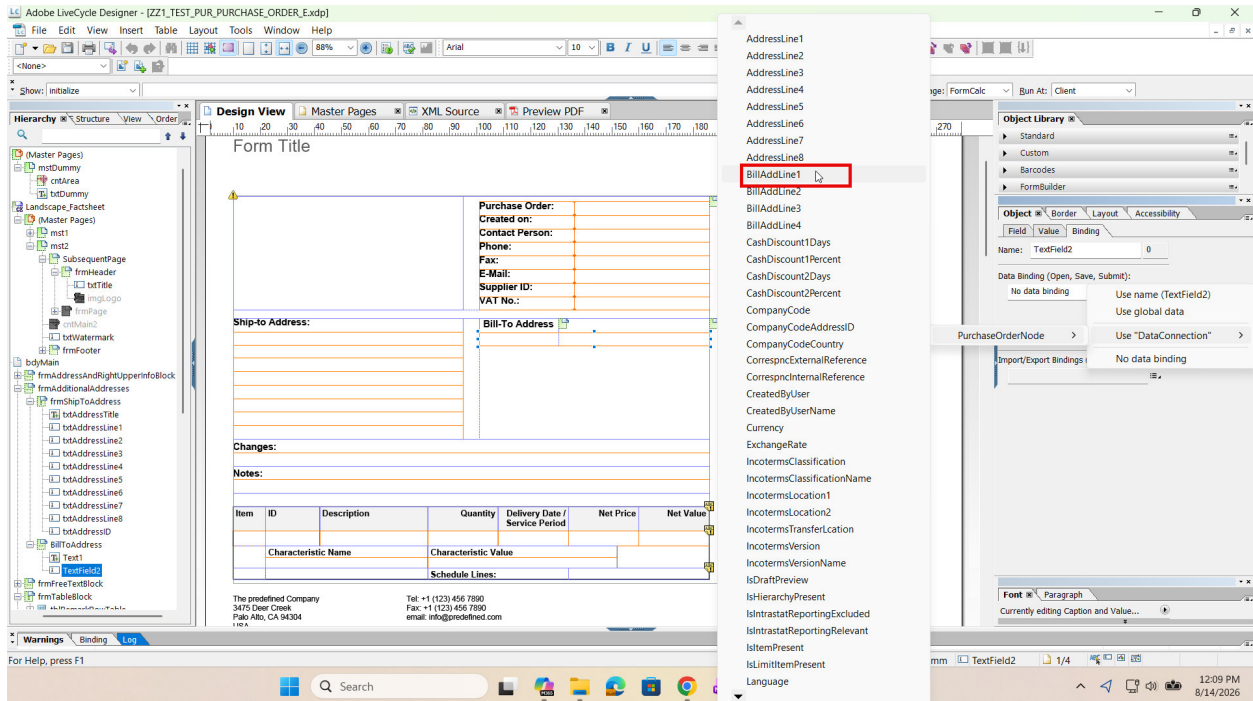
The screenshot shows the Form Designer interface with a form titled "Form Title". The form layout includes a "Purchase Order" section with fields for "Created on:", "Contact Person:", "Phone:", "Fax:", "E-Mail:", "Supplier ID:", and "VAT No.". Below this is a "Ship-to Address:" section with a "Text" field highlighted in a red box. The form also includes a "Changes:" section, a "Notes:" section, and a table with columns: "Item", "ID", "Description", "Quantity", "Delivery Date / Service Period", "Net Price", and "Net Value". The table has a header row and a body row with "Characteristic Name" and "Characteristic Value". Below the table is a "Schedule Lines:" section. The footer contains contact information for "The predefined Company" and "petco". The left sidebar shows a hierarchy of form components, and the right sidebar shows object and font properties.

Next Wrap the text added in a subform to get better control of the position where the additional fields have to be added.

The screenshot shows the Form Designer interface with a form titled "Form Title". The form layout is similar to the previous one, but with a "Bill-To Address:" section added below the "Ship-to Address:" section. A "Text Field" is highlighted in a red box within the "Bill-To Address:" section. The left sidebar shows a hierarchy of form components, and the right sidebar shows object and font properties. The "Object Library" on the right includes "Standard", "Custom", "Barcodes", and "FormBuilder". The "Object" section shows "Border" and "Layout". The "Content" section shows "Flowed" and "Allow Page Breaks within Cont". The "Presence" section shows "Visible". The "Locale" section shows "Default locale". The "Font" section shows "Paragraph".

Adjust the layout and perform binding as well. Note that the Service Object new fields are now available.

RESTRICTED

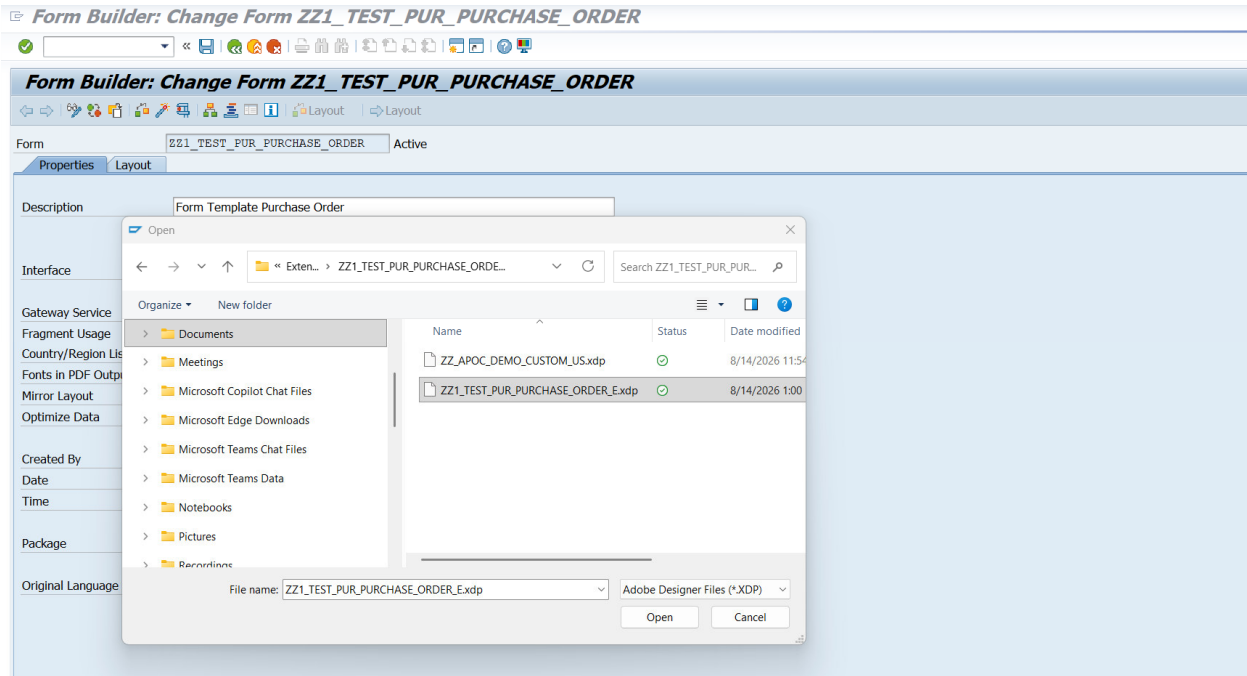


Binding and Layout. Rename the field under subform as BillToAdd1. Continue with the rest.



Upload the layout using Maintain form template.

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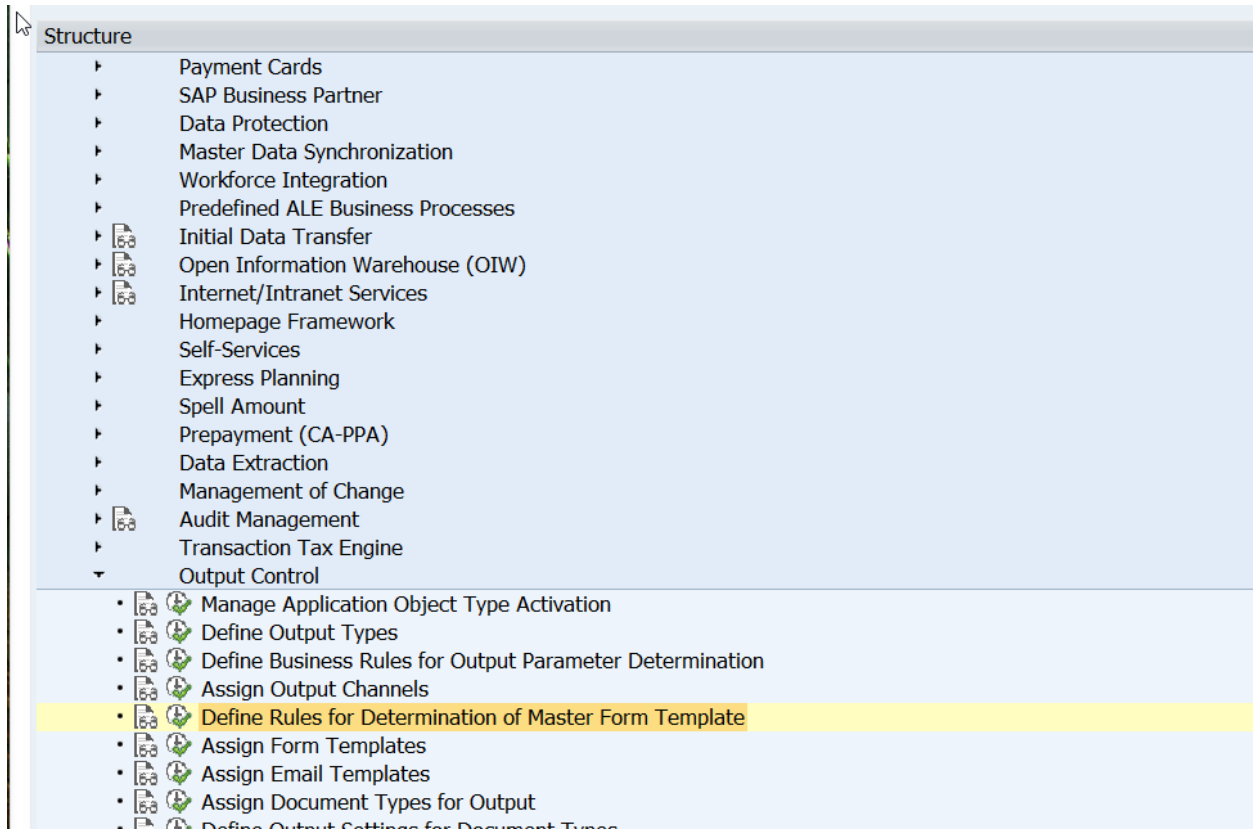
<https://community.sap.com/t5/enterprise-resource-planning-blog-posts-by-members/output-management-via-brf/ba-p/13176011>

SPRO>Cross Applications> output control> Assign Form Template and added a new purchase order with Form template id mapped to the new custom form.

The screenshot shows the 'Change View Form Template: Overview' table. The table lists various application object types and their corresponding form templates. The row for 'PURCHASE_ORDER' with 'Form Template ID' 'ZZ1_TEST_PUR_PURCHASE_ORDER' is highlighted.

Form Template	Application Object Type	Output Type	Form Type	Form Template ID	Program	Form Routine
	PURCHASE_ORDER	PURCHASE_ORDER	1 Output Forms (Gateway interfac...	MM_PUR_PURCHASE_ORDER		
	PURCHASE_ORDER	PURCHASE_ORDER	1 Output Forms (Gateway interfac...	ZMM_PUR_PURCHASE_ORDER		
	PURCHASE_ORDER	PURCHASE_ORDER	1 Output Forms (Gateway interfac...	ZZ1_TEST_PUR_PURCHASE_ORDER		
	RECONCILIATION_REPORT	RECONCILIATION_REPORT	1 Output Forms (Gateway interfac...	FI_RECONCILIATION_REPORT_RU		
	REQUEST_FOR_QUOTATION	EXTERNAL_REQUEST	1 Output Forms (Gateway interfac...	MM_PUR_RFQ_EXTERNAL		
	REQUEST_FOR_QUOTATION	INTERNAL_REQUEST	1 Output Forms (Gateway interfac...	MM_PUR_RFQ_INTERNAL		
	SALES_DOCUMENT	CASH_SALE	1 Output Forms (Gateway interfac...	SD_SLS_CASH_SALE		
	SALES_DOCUMENT	CASH_SALE_CHANGE	1 Output Forms (Gateway interfac...	SD_SLS_CASH_SALE		
	SALES_DOCUMENT	CREDIT_MEMO_REQUEST	1 Output Forms (Gateway interfac...	SD_SLS_CREDIT_MEMO_REQUEST		
	SALES_DOCUMENT	CREDIT_MEMO_REQUEST_CHANGE	1 Output Forms (Gateway interfac...	SD_SLS_CREDIT_MEMO_REQUEST		
	SALES_DOCUMENT	CUSTOMER_RETURN	1 Output Forms (Gateway interfac...	SD_SLS_CUSTOMER_RETURN		
	SALES_DOCUMENT	CUSTOMER_RETURN_CHANGE	1 Output Forms (Gateway interfac...	SD_SLS_CUSTOMER_RETURN		

I also added the master form under this setup.



RESTRICTED

Change View "Define Rules for Determination of Master Fo..": Details

Change View "Define Rules for Determination of Master Fo..": Details

New Entries

Display BC Set

Rule ID

ZNS4_0001

Define Rules for Determination of Master Form Template

Number

1

Organization Type

COMPANY

Organization ID

1710

Org. Unit Type

Org. Unit ID

Channel

PRINT

Sender Ctry/Reg.

US

Form Template

Master Form Template

ZZ_APOC_DEMO_CUSTOM_US

Content for Master Form Template

Text Type

S SAP Script

Sender Address

Footer Block 1

Footer Block 2

Footer Block 3

Footer Block 4

Footer Block 5

Footer Block 6

After the following I triggered the Purchase order using

[illegible]

